

Smooth Manifolds and Smooth Maps

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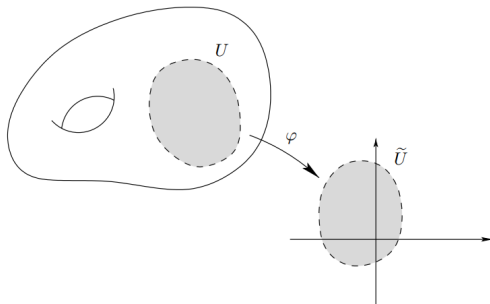
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Topological Manifolds

A topological manifold is a topological space that is

- Hausdorff
- Second countable
- Locally Euclidean
 - A topological space M is locally Euclidean of dimension n if for every point p in M , there exists a **chart** (U, ϕ) where U is open neighborhood of p and $\phi : U \rightarrow \mathbb{R}^n$ is a homeomorphism .

We call ϕ a **coordinate map** or a coordinate system on U



Topological Manifolds

Dimension of topological manifold

- For each point $p \in M$, open subset of $\mathbb{R}^n$ is not homeomorphic to an open subset of $\mathbb{R}^m$.
(*invariance of dimension*) - Prove later
- Constant on each connected component
 $M_n = \{p \in M : \text{local dimension at } p \text{ is } n\}$ is open
 $M = M_0 \sqcup M_1 \sqcup \cdots$
it is impossible for both $C \cap M_n$ and $C \cap M_m$ to be nonempty when $m \neq n$

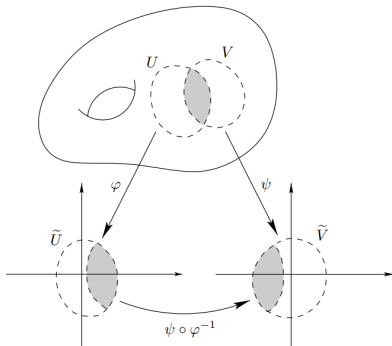
Compatible Charts

Two charts $(U, \phi : U \rightarrow \mathbb{R}^n)$ and $(V, \psi : V \rightarrow \mathbb{R}^n)$ are two charts of a topological manifold are **C^∞ -compatible** if

$$\psi \circ \phi^{-1} : \phi(U \cap V) \rightarrow \psi(U \cap V)$$

$$\phi \circ \psi^{-1} : \psi(U \cap V) \rightarrow \phi(U \cap V)$$

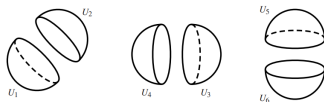
are both C^∞ .



Smooth Atlas

A C^∞ **atlas** on a topological manifold M is a collection $\mathfrak{U} = \{(U_\alpha, \phi_\alpha) : \alpha \in I\}$ of pairwise C^∞ -compatible charts that cover M .

Ex) Charts on a sphere



$$\begin{aligned} U_1 &= \{(x, y, z) \in S^2 \mid x > 0\}, & \phi_1(x, y, z) &= (y, z), \\ U_2 &= \{(x, y, z) \in S^2 \mid x < 0\}, & \phi_2(x, y, z) &= (y, z), \\ U_3 &= \{(x, y, z) \in S^2 \mid y > 0\}, & \phi_3(x, y, z) &= (x, z), \\ U_4 &= \{(x, y, z) \in S^2 \mid y < 0\}, & \phi_4(x, y, z) &= (x, z), \\ U_5 &= \{(x, y, z) \in S^2 \mid z > 0\}, & \phi_5(x, y, z) &= (x, y), \\ U_6 &= \{(x, y, z) \in S^2 \mid z < 0\}, & \phi_6(x, y, z) &= (x, y). \end{aligned}$$

$$\phi_1 \circ \phi_3^{-1}(x, z) = (\sqrt{1 - x^2 - z^2}, z)$$

is C^∞ on $\phi_3(U_1 \cap U_3)$

Lemma 5.8

Let $\{(U_\alpha, \phi_\alpha)\}$ be an atlas on a topological manifold. If two charts (V, ψ) and (W, σ) are both compatible with the atlas $\{(U_\alpha, \phi_\alpha)\}$, then they are compatible with each other.

Proof.

Let $p \in V \cap W$. $p \in U_\alpha$ for some α .

Then $\sigma \circ \psi^{-1} = (\sigma \circ \phi_\alpha^{-1}) \circ (\phi_\alpha \circ \psi^{-1})$ holds on $\psi(V \cap W \cap U_\alpha)$ and C^∞ .

Therefore, $\sigma \circ \psi^{-1}$ is C^∞ at $\psi(p)$. □

Proposition 5.10

Any atlas $\mathfrak{U} = \{(U_\alpha, \phi_\alpha)\}$ on a topological manifold is contained in a unique maximal atlas.

Proof.

Enlarge the atlas $\mathfrak{U}$ by adding all charts compatible with $\mathfrak{U}$. By Lemma 5.8, all charts are compatible with one another. So the enlarged collection of charts is an atlas.; Maximal
It contains all charts that are compatible with $\mathfrak{U}$; Unique □

Smooth Manifolds

Definition

A **smooth or C^∞ manifold** is a topological manifold M together with a maximal atlas (differentiable structure on M).

To show that a topological space M is a C^∞ manifold, it suffices to check that

- M is a Hausdorff and second countable
- M has a C^∞ atlas (not necessarily maximal)

Smooth Manifolds

Remark

A topological manifold can admit distinct differentiable structures.

Let $\mathbb{R}$ be the real line with the differentiable structure given by the maximal atlas of the chart $(\mathbb{R}, id_{\mathbb{R}})$, and let $\mathbb{R}'$ be the real line with the differentiable structure given by the maximal atlas of the chart $(\mathbb{R}, \phi)$ where $\phi(x) = x^{1/3}$.

But $\phi \circ id_{\mathbb{R}}^{-1}(x) = x^{1/3}$ is not C^∞ at $x = 0$. Therefore, two charts are not compatible and two differentiable structures are distinct.

Examples of Smooth Manifolds

- **Open subset of a manifold**

Any open subset V of a manifold M is also a manifold.

- If $\{(U_\alpha, \phi_\alpha)\}$ is an atlas for M , then $\{(U_\alpha \cap V, \phi_\alpha|_{U_\alpha \cap V})\}$ is an atlas for V .
: Each $(U_\alpha \cap V, \phi_\alpha|_{U_\alpha \cap V})$ is still a chart, and the transition map $\phi_\alpha|_{U_\alpha \cap V} \circ \phi_\beta|_{U_\beta \cap V}^{-1}$ is C^∞ on $\phi_\beta(U_\alpha \cap U_\beta \cap V)$.
- V is Hausdorff and second countable.
 - $x, y \in V$, there exist disjoint open $x \in A, y \in B$ such that $x \in V \cap A$ and $y \in V \cap B$
 - $\{B_n\}_{n \in \mathbb{N}}$: countable basis of M . Then $\{B_n \cap V\}_{n \in \mathbb{N}}$: countable basis of V .

Examples of Smooth Manifolds

• Product manifold

- If $\{(U_\alpha, \phi_\alpha)\}$ and $\{(V_\beta, \psi_\beta)\}$ are C^∞ atlases for the manifolds M and N of dimensions m and n , respectively, then the collection

$$\{(U_\alpha \times V_\beta, \phi_\alpha \times \psi_\beta)\}$$

of charts is a C^∞ atlas on $M \times N$, where $(f \times g)(x, y) = (f(x), g(y))$.

: Each $(U_\alpha \times V_\beta, \phi_\alpha \times \psi_\beta)$ is still a chart, and the transition map

$$(\phi_\alpha \times \psi_\beta) \circ (\phi_{\alpha'} \times \psi_{\beta'})^{-1} = (\phi_\alpha \circ \phi_{\alpha'}^{-1}) \times (\psi_\beta \circ \psi_{\beta'}^{-1})$$

is C^∞ on $\phi_{\alpha'}(U_\alpha \cap U_{\alpha'}) \times \psi_{\beta'}(V_\beta \cap V_{\beta'})$.

- $M \times N$ is Hausdorff and second countable.
 - $(x_1, y_1), (x_2, y_2) \in M \times N$ with $x_1 \neq x_2$, there exist disjoint open $A_1, A_2 \subset M$ such that $x_1 \in A_1, x_2 \in A_2$. Then $A_1 \times N$ and $A_2 \times N$ separate (x_1, y_1) and (x_2, y_2) .
 - $\{B_i\}_{i \in \mathbb{N}}$: countable basis of M , and $\{C_j\}_{j \in \mathbb{N}}$: countable basis of N . Then $\{B_i \times C_j\}_{i, j \in \mathbb{N}}$: countable basis of $M \times N$.

Therefore, $M \times N$ is a C^∞ manifold of dimension $m + n$.

Smooth Maps on a Manifold

Definition

Let M be a smooth manifold of dimension n .

A function $f : M \rightarrow \mathbb{R}$ is said to be C^∞ or **smooth** at a point p in M if there is a chart (U, ϕ) about p such that

$$f \circ \phi^{-1} : \phi(U) (\subset \mathbb{R}^n) \rightarrow \mathbb{R}$$

is C^∞ at $\phi(p)$.

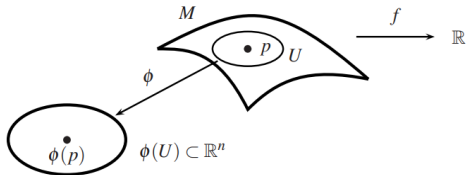


Fig. 6.1. Checking that a function f is C^∞ at p by pulling back to $\mathbb{R}^n$.

Smoothness Independent to chart

Smoothness of a map $f : M \rightarrow \mathbb{R}$ at a point is **independent** of the chart (U, ϕ)

For another chart (V, ψ) about $p \in M$, then on $\psi(U \cap V)$,

$$f \circ \psi^{-1} = (f \circ \phi^{-1}) \circ (\phi \circ \psi^{-1})$$

$\phi \circ \psi^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is C^∞ : compatibility of charts

Smooth Maps Between Manifolds

Definition

Let N and M be manifolds of dimension n and m , respectively.

A continuous map $F : N \rightarrow M$ is C^∞ at a point $p \in N$ if there are charts (U, ϕ) about $p \in N$ and (V, ψ) about $F(p) \in M$ such that

$$\psi \circ F \circ \phi^{-1} : \phi(U \cap F^{-1}(V)) \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$$

is C^∞ at $\phi(p)$.

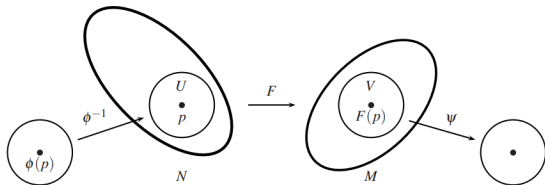


Fig. 6.3. Checking that a map $F: N \rightarrow M$ is C^∞ at p .

Smoothness Independent to chart

Smoothness of a map $F : N \rightarrow M$ at a point is **independent** of the charts (U, ϕ) and (V, ψ)

For other charts (U', ϕ') about $p \in N$ and (V', ψ') about $F(p) \in M$,

$$\psi \circ F \circ \phi^{-1} = (\psi \circ \psi'^{-1}) \circ (\psi' \circ F \circ \phi'^{-1}) \circ (\phi' \circ \phi^{-1})$$

is C^∞ at $\phi(p)$.

Smooth Maps Between Manifolds

Smoothness of a map in terms of charts

Let N and M be manifolds of dimension n and m , and $F : N \rightarrow M$ a continuous map. The following are equivalent:

- (i) The map $F : N \rightarrow M$ is C^∞ .
- (ii) There exist atlases $\mathfrak{U}$ for N and $\mathfrak{V}$ for M such that for every chart $(U, \phi) \in \mathfrak{U}$ and $(V, \psi) \in \mathfrak{V}$, the map

$$\psi \circ F \circ \phi^{-1} : \phi(U \cap F^{-1}(V)) \rightarrow \mathbb{R}^m$$

is C^∞ .

- (iii) For every chart (U, ϕ) on N and (V, ψ) on M , the map

$$\psi \circ F \circ \phi^{-1} : \phi(U \cap F^{-1}(V)) \rightarrow \mathbb{R}^m$$

is C^∞ .

Smooth Maps Between Manifolds

Smoothness of a map in terms of charts

Proof.

(ii) $\rightarrow$ (i) $p \in N$, there exist $(U, \phi) \in \mathfrak{U}$ about p and $(V, \psi) \in \mathfrak{V}$ about $F(p)$ such that $\psi \circ F \circ \phi^{-1}$ is C^∞ at $\phi(p)$.

(i) $\rightarrow$ (iii) Let $p \in U \cap F^{-1}(V)$. Then (U, ϕ) and (V, ψ) are charts about p and $F(p)$, respectively. Then $\psi \circ F \circ \phi^{-1}$ is C^∞ at $\phi(p)$.

(iii) $\rightarrow$ (ii) Trivial

Smooth Maps Between Manifolds

Composition of smooth maps

If $F : N \rightarrow M$ and $G : M \rightarrow P$ are C^∞ maps of manifolds, then the composite $G \circ F : N \rightarrow P$ is C^∞ .

Proof.

Let (U, ϕ) , (V, ψ) , and (W, σ) be charts on N , M , and P , respectively. Then

$$\sigma \circ (G \circ F) \circ \phi^{-1} = (\sigma \circ G \circ \psi^{-1}) \circ (\psi \circ F \circ \phi^{-1}).$$

$\sigma \circ G \circ \psi^{-1}$ and $\psi \circ F \circ \phi^{-1}$ are C^∞ maps of open subsets of Euclidean spaces. □

Diffeomorphism

Definition

A **diffeomorphism** of manifolds is a bijective C^∞ map $F : N \rightarrow M$ whose inverse F^{-1} is also C^∞ .

Proposition 6.10

If (U, ϕ) is a chart on a manifold M of dimension n , then the coordinate map $\phi : U \rightarrow \phi(U) \subset \mathbb{R}^n$ is a diffeomorphism.

Proof. We use the atlas $\{(U, \phi)\}$ with a single chart on U and the atlas $\{(\phi(U), 1_{\phi(U)})\}$ with a single chart on $\phi(U)$. Since

$$1_{\phi(U)} \circ \phi \circ \phi^{-1} : \phi(U) \rightarrow \phi(U)$$

is the identity map, it is C^∞ . Therefore, ϕ is C^∞ . Using

$$\phi \circ \phi^{-1} \circ 1_{\phi(U)} = 1_{\phi(U)} : \phi(U) \rightarrow \phi(U),$$

ϕ^{-1} is also C^∞ . □

Diffeomorphism

Proposition 6.11

Let U be an open subset of a manifold M of dimension n . If $F : U \rightarrow F(U) \subset \mathbb{R}^n$ is a diffeomorphism onto an open subset of $\mathbb{R}^n$, then (U, F) is a chart in the differentiable structure of M .

Proof. For any chart (U_α, ϕ_α) in the maximal atlas of M , both ϕ_α and ϕ_α^{-1} are C^∞ by Proposition 6.10. As composites of C^∞ maps, both

$$F \circ \phi_\alpha^{-1} \quad \text{and} \quad \phi_\alpha \circ F^{-1}$$

are C^∞ . Hence, (U, F) is compatible with the maximal atlas. By the maximality of the atlas, the chart (U, F) is in the atlas. □

By Proposition 6.10 and 6.11, coordinate maps are diffeomorphisms, and conversely, every diffeomorphism of an open subset of a manifold with an open subset of a Euclidean space can serve as a coordinate map.

Examples of Smooth Maps

- **Projection map**

Let M and N be manifolds of dimension m, n and $\pi : M \times N \rightarrow M$, $\pi(p, q) = p$.
: $(p, q) \in M \times N$. Let $(U, \phi) = (U, x^1, \dots, x^m)$ and $(V, \psi) = (V, y^1, \dots, y^n)$ be charts about p and q respectively. Then $(U \times V, \phi \times \psi)$ is a chart about (p, q) , (product manifold)

$$(\phi \circ \pi \circ (\phi \times \psi)^{-1})(a^1, \dots, a^m, b^1, \dots, b^n) = (a^1, \dots, a^m),$$

is a C^∞ map from $(\phi \times \psi)(U \times V) \subset \mathbb{R}^{m+n}$ to $\phi(U) \subset \mathbb{R}^m$. Therefore π is C^∞ at (p, q) . Since (p, q) was an arbitrary point in $M \times N$, π is C^∞ on $M \times N$. $\square$

Examples of Smooth Maps

- **Cartesian product**

Let M_1, M_2 and N be manifolds of dimension m_1, m_2, n and $F = (f_1, f_2) : N \rightarrow M_1 \times M_2$ is C^∞ if and only if $f_i : N \rightarrow M_i, i = 1, 2$ are both C^∞ .
($\Rightarrow$) smoothness of projection map and composition
($\Leftarrow$) $p \in N$. Let $(U, \phi), (V_1, \psi_1)$ and (V_2, ψ_2) be charts about $p, f_1(p)$ and $f_2(p)$ respectively.

$$((\psi_1 \times \psi_2) \circ F \circ \phi^{-1}) = ((\psi_1 \circ F \circ \phi^{-1}), (\psi_2 \circ F \circ \phi^{-1}))$$

is a C^∞ map from an open subset of $\mathbb{R}^n$ to an open subset of $\mathbb{R}^{m_1+m_2}$. Therefore F is C^∞ at p . Since p was an arbitrary point in N , F is C^∞ on N . □

Partial Derivatives

On a manifold M of dimension n , let $(U, \phi) = (U, x^1, \dots, x^n)$ be a chart and $f : M \rightarrow \mathbb{R}$ a C^∞ map.

For $p \in U$, the **partial derivative** $\partial f / \partial x^i$ of f with respect to x^i at p is defined as

$$\left. \frac{\partial}{\partial x^i} \right|_p f = \frac{\partial f}{\partial x^i}(p) := \frac{\partial(f \circ \phi^{-1})}{\partial r^i}(\phi(p)) = \left. \frac{\partial}{\partial r^i} \right|_{\phi(p)} (f \circ \phi^{-1})$$

where $r^1, \dots, r^n$ are the standard coordinates on $\mathbb{R}^n$, therefore $x^i = r^i \circ \phi$.

Jacobian matrix

If $F : N \rightarrow M$ be a smooth map, and let $(U, \phi) = (U, x^1, \dots, x^n)$ and $(V, \psi) = (V, y^1, \dots, y^m)$ be charts on N and M , respectively, such that $F(U) \subset V$. Denote by

$$F^i := y^i \circ F = r^i \circ \psi \circ F : U \rightarrow \mathbb{R}.$$

i -th component of F in the chart (V, ψ) . Then the matrix $[\partial F^i / \partial x^j]$ is the **Jacobian matrix** of F relative to the charts (U, ϕ) and (V, ψ) .

Inverse Function Theorem

$F : N \rightarrow M$ is **locally invertible** or a **local diffeomorphism** at $p \in N$ if p has a neighborhood U on which $F|_U : U \rightarrow F(U)$ is a diffeomorphism.

Inverse function theorem for $\mathbb{R}^n$

Let $F : W(\subset \mathbb{R}^n) \rightarrow \mathbb{R}^n$ be a C^∞ map defined on an open subset W . For any point $p \in W$, the map F is locally invertible at p if and only if

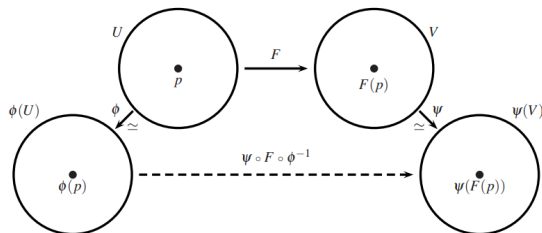
$$\det \left[\frac{\partial F^i}{\partial x^j}(p) \right] \neq 0.$$

Inverse function theorem for manifolds

Let $F : N \rightarrow M$ be a C^∞ map between two manifolds of the same dimension. Let $(U, \phi) = (U, x^1, \dots, x^n)$ and $(V, \psi) = (V, y^1, \dots, y^n)$ be charts about p and $F(p)$, $F(U) \subset V$. Set $F^i = y^i \circ F$. Then F is locally invertible at p if and only if

$$\det \left[\frac{\partial F^i}{\partial x^j}(p) \right] \neq 0.$$

Inverse Function Theorem



Proof. Since $F^i = y^i \circ F = r^i \circ \psi \circ F$, the Jacobian matrix of F relative to the charts (U, ϕ) and (V, ψ) is

$$\left[\frac{\partial F^i}{\partial x^j}(p) \right] = \left[\frac{\partial (r^i \circ \psi \circ F)}{\partial x^j}(p) \right] = \left[\frac{\partial (r^i \circ \psi \circ F \circ \phi^{-1})}{\partial r^j}(\phi(p)) \right],$$

which is precisely the Jacobian matrix at $\phi(p)$ of the map

$$\psi \circ F \circ \phi^{-1} : \phi(U)(\subset \mathbb{R}^n) \rightarrow \psi(V)(\subset \mathbb{R}^n)$$

between two open subsets of $\mathbb{R}^n$.

Inverse Function Theorem

By the inverse function theorem for $\mathbb{R}^n$,

$$\det \left[\frac{\partial F^i}{\partial x^j}(p) \right] = \det \left[\frac{\partial (r^i \circ (\psi \circ F \circ \phi^{-1}))}{\partial r^j}(\phi(p)) \right] \neq 0$$

if and only if $\psi \circ F \circ \phi^{-1}$ is locally invertible at $\phi(p)$. Since ϕ and ψ are diffeomorphisms, $\square$

Inverse Function Theorem

Corollary Let N be a manifold of dimension n . A set of n smooth functions $F^1, \dots, F^n : N \rightarrow \mathbb{R}$ defined on a coordinate neighborhood $(U, x^1, \dots, x^n)$ of a point $p \in N$ forms a coordinate system about p if and only if $\det \left[\frac{\partial F^i}{\partial x^j}(p) \right] \neq 0$.

Proof. Let $F = (F^1, \dots, F^n) : U \rightarrow \mathbb{R}^n$.

Then $\det \left[\frac{\partial F^i}{\partial x^j}(p) \right] \neq 0$ if and only if $F : U \rightarrow \mathbb{R}^n$ is locally invertible at p , by the inverse function theorem. This is equivalent to the existence of a neighborhood W of p in N such that

$$F : W \rightarrow F(W)$$

is a diffeomorphism, by the definition of local invertibility.

By Proposition 6.11, this is equivalent to saying that

$$(W, F^1, \dots, F^n)$$

is a coordinate chart about p in the differentiable structure of N . □

Inverse Function Theorem

Example

Find all points in $\mathbb{R}^3$ in a neighborhood of which the functions $F = (x, x^2 + y^2 + z^2 - 1, z)$ can serve as a local coordinate system.

$$\det \left[\frac{\partial (r^i \circ (\psi \circ F \circ \phi^{-1}))}{\partial r^j} (\phi(p)) \right] = \det \begin{pmatrix} 1 & 0 & 0 \\ 2x & 2y & 2z \\ 0 & 0 & 1 \end{pmatrix} = 2y$$

F can serve as a local coordinate system about points in $\{(x, y, z) : y \neq 0\}$.